Graph Signal Processing and Interpolation for EEG Channel Reconstruction

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***Abstract*—This paper studies EEG channel reconstruction under missing-channel conditions using a graph signal processing (GSP) benchmark with reproducible execution artifacts. Following the GSP perspective popularized by Ortega and collaborators, we model EEG electrodes as graph vertices and evaluate reconstruction quality through both spatial and temporal criteria. We compare geometric baselines, graph-based estimators, and graph-time regularization methods under scenario-driven masking protocols. The current B2 stage includes full-scale batched runs and consolidated publication tables with MAE, RMSE, DTW, and SNR statistics. Results indicate that graph-aware and TV/time methods remain consistently competitive, while ranking differences are scenario-dependent and therefore require variance-aware interpretation. We also report the current BGSRP status transparently: strong proxy performance in Python, with strict MATLAB/GSPBox 1:1 equivalence still pending.**

***Index Terms*—EEG reconstruction, graph signal processing, interpolation, missing channels, temporal regularization**

I. Introduction

Electroencephalography (EEG) recordings frequently exhibit missing channels due to contact issues, amplifier saturation, movement artifacts, or quality-control rejection. In practical studies, this is not a minor inconvenience: missing channels can distort spatial interpretations, degrade biomarkers, and reduce the reliability of downstream decoding pipelines. Because of that, channel interpolation has been studied for decades, from classical spatial interpolation to graph-based reconstruction.

Graph Signal Processing (GSP) provides a natural framework for this problem by representing electrodes as graph vertices and modeling inter-channel relationships through weighted edges. The general perspective is consistent with the graph-signal literature associated with Ortega and collaborators [**?**], [**?**], where smoothness, bandlimitedness, and spectral structure are used to replace purely geometric interpolation with graph-constrained estimation.

Even though many methods have been proposed in the literature [**?**], [**?**], [**?**], in practice it is still difficult to compare them fairly in a single reproducible pipeline. Different papers use different masks, datasets, and evaluation conventions, so conclusions are often hard to transfer to real EEG workflows. This thesis-paper track tries to reduce that gap with a unified benchmark and publication-ready artifacts.

The main contributions of this manuscript are:

- A unified and reproducible pipeline that covers graph construction, missing-channel simulation, reconstruction, and quantitative evaluation.
- A cross-family comparison including geometric baselines, graph-based interpolation, and temporal regularization methods.
- A publication-oriented artifact set with frozen configurations, raw outputs, and summary tables.

At the current B2/B3/B4 stage, we evaluate performance using MAE, RMSE, DTW, and SNR under scenario-driven masking conditions (4 scenarios, 10%, 20%, 30% y 40% missing) with repeated runs and consolidated variability statistics.

This is still an evolving manuscript, so we intentionally keep a transparent scope statement: most of the benchmark is consolidated, but strict BGSRP equivalence against the original MATLAB/GSPBox environment remains a documented limitation rather than a completed claim.

II. Related Work

Graph-based interpolation for missing data in structured signals is well established in the signal processing on graphs literature. Early formulations by Narang and Ortega [**?**] showed that interpolation can be cast as a graph-constrained reconstruction problem, while later work by Puy *et al.* [**?**] clarified when bandlimited graph signals can be recovered from partial observations. In parallel, the broader GSP framing summarized by Shuman *et al.* [**?**] and later graph-wavelet constructions by Hammond *et al.* [**?**] helped establish the intuition that graph structure can replace purely Euclidean interpolation when the data lives on irregular domains.

More recent methods include RKHS-based recovery schemes such as BGSRP [**?**], as well as time-varying formulations that combine spatial and temporal smoothness [**?**]. In EEG settings, these ideas are especially relevant because electrodes are spatially arranged, the geometry is not perfectly uniform, and physiological activity evolves over time in a correlated but not trivial way. For this reason, methods that exploit both graph smoothness and temporal regularity are often more appropriate than simple baselines when channels are missing.

The present work does not attempt to introduce a new interpolation theory. Instead, it tries to make the comparison fairer and more reproducible by evaluating these families under shared data, masking, and metric protocols, using the same EEG benchmark pipeline and the same consolidated evidence artifacts.

III. Methods

We represent EEG channels as graph vertices and model inter-channel relationships with weighted adjacency matrices. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{W})$ be an undirected weighted graph where $\mathcal{V}$

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is the set of N electrode vertices, $\mathcal{E}$ the edge set, and $\mathbf{W} \in \mathbb{R}^{N \times N}$ the symmetric weight matrix. The combinatorial graph Laplacian is $\mathbf{L} = \mathbf{D} - \mathbf{W}$, where $\mathbf{D}$ is the degree matrix. A graph signal $\mathbf{x} \in \mathbb{R}^N$ assigns a scalar value to each electrode at a given time instant.

The goal is not to propose a new theory, but to compare 15 instant and 10 TV/time methods under the same experimental conditions and with the same evaluation pipeline.

A. Graph Construction

We evaluate 10 graph constructors organized into three categories:

- *Neighborhood graphs*: k-NN (`knn`), Gaussian-weighted k-NN (`knng`), variable-k NN (`vknng`), epsilon-ball (`epsilon_ball`), and minimum spanning tree (`mst`).
- *Dense/global graphs*: fully connected inverse-distance (`fully_connected_inverse_distance`) and Gaussian kernel (`gaussian`).
- *Adaptive/learned graphs*: non-negative kernel regression [?] (`nnk`), adaptive edge weighting [?] (`aew`), and graph learning via log-degree regularization [?] (`kalofolias`).

In phase B2/B3/B4, we retained only constructors with demonstrated numerical stability across the full-scale batched runs.

B. Interpolation Families

Let $\mathcal{K}$ denote the set of known (observed) channels and $\mathcal{U}$ the unknown (missing) channels. We compare three families:

a) Instant methods: These operate independently at each time sample. The graph Laplacian regularization problem takes the form:

$$\hat{\mathbf{x}}_{\mathcal{U}} = \arg\min_{\mathbf{z}} \|\mathbf{z} - \mathbf{x}_{\mathcal{K}}\|^2 + \alpha\, \mathbf{z}^\top \mathbf{L}_{\mathcal{U}\mathcal{U}} \mathbf{z}, \tag{1}$$

yielding the closed-form solution $\hat{\mathbf{x}}_{\mathcal{U}} = -\mathbf{L}_{\mathcal{U}\mathcal{U}}^{-1} \mathbf{L}_{\mathcal{U}\mathcal{K}} \mathbf{x}_{\mathcal{K}}$ for the pure GSP case ($\alpha \to \infty$, i.e., `gsp`), and the regularized variant for `tikhonov`. Geometric baselines (IDW, splines, RBF) do not use the graph. The BGSRP method [?] extends this to a bandlimited RKHS recovery.

b) TV/time methods: These jointly optimize spatial and temporal criteria over a signal matrix $\mathbf{X} \in \mathbb{R}^{N \times T}$:

$$\hat{\mathbf{X}} = \arg\min_{\mathbf{Z}} \|\mathbf{Z}_{\mathcal{K},:} - \mathbf{X}_{\mathcal{K},:}\|_F^2 + \alpha \operatorname{tr}(\mathbf{Z}^\top \mathbf{L} \mathbf{Z}) + \beta \|\Delta_t \mathbf{Z}\|_F^2, \tag{2}$$

where Δ_t is a finite-difference operator along the time axis [?]. This covers `trss`, `sobolev_temporal`, `tv`, and `temporal_laplacian`.

c) Geometric/spline baselines: Classical alternatives that do not exploit the graph: linear, nearest-neighbor, spherical spline [?], RBF thin-plate spline, and surface spline.

C. Evaluation Metrics

Performance is assessed with four complementary metrics: mean absolute error (MAE), root mean square error (RMSE), dynamic time warping distance (DTW), and signal-to-noise ratio (SNR). We report mean $\pm$ standard deviation and approximate 95% confidence intervals from repeated runs.

The four-metric combination is essential in EEG: MAE and RMSE capture pointwise amplitude accuracy, DTW penalizes temporal misalignment even under small shifts, and SNR provides a global quality view. A method with low MAE may still distort temporal dynamics, and a method competitive on DTW may not rank first on MAE.

D. Statistical Analysis

Family-level differences are tested with Wilcoxon paired tests (STAT-02, $\alpha = 0.05$) over shared scenario-contexts. Method-level pairwise contrasts use Mann-Whitney U. All statistical tests are computed on the B2 full-scale batched raw outputs prior to mean aggregation.

IV. Experimental Setup

Experiments follow a fixed pipeline: data loading, graph construction, missing-channel simulation, reconstruction, and quantitative evaluation.

For B2, we ran the benchmark in batches and then consolidated the outputs into publication-oriented artifacts. The implementation uses the scripts `experiments/run_b2_batched.py` and `experiments/consolidate_b2_publication.py`.

A. Masking Protocol

Masking includes scenario-driven patterns and missing ratios of 0.10, 0.20, 0.30, and 0.40. In the current stage we use realistic region-based scenarios (frontal band, occipital band, temporal lateral, and midline central) over the available EEG sample set.

B. Reproducibility

All scripts and output artifacts are tracked in versioned folders and frozen configurations. The most relevant B2 outputs are:

- `results/opt_benchmark_b2_full_scale_summary.csv`
- `results/b2_publication_ranking_final.csv`
- `results/b2_publication_topk_by_family_scenario.`
- `results/b2_publication_bgsrp_gap_residual.csv`